# Measure Theory and Lebesgue Integration

I

Anqiao Ouyang

August 2026

## Introduction

This article introduces the foundations of measure theory and the Lebesgue integral, which together form the backbone of modern real analysis. We begin with $\sigma$-algebras and measures, construct the Lebesgue measure on $\mathbb{R}^n$, and develop the integral with respect to a general measure.

## 1 Sigma-Algebras and Measurable Spaces

Let $X$ be a non-empty set. A collection $\mathcal{F} \subseteq \mathcal{P}(X)$ is a *$\sigma$-algebra* on $X$ if:

(i) $X \in \mathcal{F}$;

(ii) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;

(iii) if $(A_n)_{n=1}^{\infty}$ is a sequence in $\mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

The pair $(X, \mathcal{F})$ is called a *measurable space*.

> The Borel $\sigma$-algebra $\mathcal{B}(\mathbb{R})$ is the $\sigma$-algebra generated by the open sets of $\mathbb{R}$. It contains all open sets, closed sets, countable intersections of open sets ($G_\delta$ sets), and countable unions of closed sets ($F_\sigma$ sets).

## 2 Measures

Let $(X, \mathcal{F})$ be a measurable space. A function $\mu \colon \mathcal{F} \to [0, \infty]$ is a *measure* if:

(i) $\mu(\varnothing) = 0$;

(ii) (countable additivity) for any sequence $(A_n)$ of pairwise disjoint sets in $\mathcal{F}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

Let $\mu_0$ be a pre-measure on an algebra $\mathcal{A} \subseteq \mathcal{P}(X)$. Then there exists a measure $\mu$ on $\sigma(\mathcal{A})$ that extends $\mu_0$. If $\mu_0$ is $\sigma$-finite, the extension is unique.

***Proof.*** Define the outer measure $\mu^*(E) = \inf \sum_n \mu_0(A_n)$, where the infimum is taken over all countable covers $\{A_n\} \subseteq \mathcal{A}$ of $E$. One then shows that the collection of $\mu^*$-measurable sets forms a $\sigma$-algebra containing $\mathcal{A}$, and $\mu^*$ restricted to this $\sigma$-algebra is a measure extending $\mu_0$. □

## 3 The Lebesgue Integral

Let $(X, \mathcal{F}, \mu)$ be a measure space and $\varphi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ a non-negative simple function. The Lebesgue integral of $\varphi$ is

$$\int_X \varphi \, d\mu = \sum_{i=1}^{n} a_i \, \mu(A_i).$$

Let $(f_n)$ be a sequence of non-negative measurable functions with $f_n \uparrow f$ pointwise. Then

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

The Monotone Convergence Theorem allows us to exchange limits and integrals for increasing sequences. This is the engine that powers most constructions of the Lebesgue integral: approximate from below by simple functions and pass to the limit.

Let $(f_n)$ be a sequence of measurable functions converging pointwise to $f$. If there exists an integrable function $g$ such that $|f_n| \leq g$ for all $n$, then

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

The domination condition $|f_n| \leq g$ with $g$ integrable cannot be dropped. The sequence $f_n = n \cdot \mathbf{1}_{(0,1/n)}$ converges pointwise to 0 on $(0, 1)$, yet $\int f_n = 1$ for all $n$.